

Mutagenic and antimutagenic effects of methanol extracts of unfermented and fermented black soybeans.

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In this study, solid fermentation of steamed black soybean with various GRAS (Generally recognized as safe) filamentous-fungi including *Aspergillus awamori*, *Aspergillus oryzae* BCRC 30222, *Aspergillus sojae* BCRC 30103, *Rhizopus azygosporus* BCRC 31158 and *Rhizopus* sp. No. 2 was performed. Mutagenicity and antimutagenicity of the methanol extracts of unfermented and fermented steamed black soybeans against 4-nitroquinoline-N-oxide (4-NQO), a direct mutagen and Benzo[a]pyrene (B[a]P), an indirect mutagen, on *Salmonella Typhimurium* TA100 and TA 98, were examined. The methanol extracts of unfermented and fermented steamed black soybeans show no mutagenic activity for either test strains at the doses tested. The extracts inhibited mutagenesis by either 4-NQO or B[a]P in *S. Typhimurium* TA100 and TA98. Fermentation with fungi also enhanced the antimutagenic effect of black soybean while the antimutagenic effect of the fermented black soybeans extract varied with the starter organism, mutagen, and test strain of *S. Typhimurium* examined. Generally, the extracts of *A. awamori*-fermented black soybean exhibited the highest antimutagenic effect. With strain TA100, the inhibitory effects of 5.0 mg of *A. awamori*-fermented black soybean extract per plate on the mutagenic effects of 4-NQO and B[a]P were 92% and 89%, respectively, while the corresponding rates for extract of unfermented were 41% and 63%, respectively. With strain 98, the inhibition rates were 94 and 81% for the fermented bean extract and 58% and 44% for the unfermented bean extracts. Testing of extracts prepared from black soybean by *A. awamori* at temperatures 25, 30 and 35 degrees C and for times of 1-5 days revealed that, generally, the extract prepared from beans fermented at 30 degrees C for 3 days exhibited the greatest inhibition against the mutagenic effects of 4-NQO and B[a]P.